REVISION WORKSHEET

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الإمارات العربية المتحدة
وزارة التربية والتعليم

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Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 12 Advanced | Module 10 Lesson 2 | The Many Forms of Energy

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: KE = ½mv²; KErot = ½Iω²; GPE = mgh; PEspring = ½kx²; Erest = mc²; g = 9.8 m/s².

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 14–22.

P01. Translational kinetic energy is proportional to:

A. Speed only B. The inverse of mass
C. The square of speed D. The square of time in every situation

Answer

C. The square of speed.

P02. Gravitational potential energy depends on the chosen:

A. Coordinate unit only with no conversion B. Object colour
C. Direction of horizontal travel only D. Zero-height reference

Answer

D. Zero-height reference.

P03. Elastic potential energy can be stored in a:

A. Stretched spring B. Stationary uncompressed spring at its defined minimum only
C. Reference point with no object D. Unit conversion

Answer

A. Stretched spring.

P04. Rest energy is related to mass by:

A. $E = m/c^2$ B. $E = mc^2$
C. $E = mv$ D. $E = mg$ only

Answer

B. $E = mc^2$.

P05. For the same I, doubling angular speed makes rotational kinetic energy:

A. Twice as large B. Half as large
C. Four times as large D. Unchanged

Answer

C. Four times as large.

P06. A 4 kg object moves at 3.0 m/s. Find its translational kinetic energy.

A. 18 J B. 36 J
C. 9 J D. -18 J

Answer

A. 18 J. $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 4 \times 3.0^2 = 18$ J.

P07. A 4 kg object is 3.0 m above the chosen zero level. Find its gravitational potential energy. Use $g = 9.8$ m/s².

A. 235.2 J B. 117.6 J
C. 58.8 J D. -117.6 J

Answer

B. 117.6 J. $GPE = mgh = 4 \times 9.8 \times 3.0 = 117.6$ J.

P08. A rotor has moment of inertia 0.4 kg m² and angular speed 5.0 rad/s. Find its rotational kinetic energy.

A. 10 J B. 2.5 J
C. 5 J D. -5 J

Answer

C. 5 J. $KE_{rot} = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 0.4 \times 5.0^2 = 5$ J.

P09. A spring with constant 200 N/m is stretched 0.100 m from equilibrium. Find its elastic potential energy using $PE = \frac{1}{2}kx^2$.

A. 2 J B. 0.5 J
C. -1 J D. 1 J

Answer

D. 1 J. $PE = \frac{1}{2}kx^2 = \frac{1}{2} \times 200 \times 0.100^2 = 1$ J.

P10. Find the rest energy of mass 2×10^{-9} kg. Use $E = mc^2$ and $c = 3.00 \times 10^8$ m/s.

A. 1.8×10^8 J B. 3.6×10^8 J
C. 9×10^7 J D. -1.8×10^8 J

Answer

A. 1.8×10^8 J. $E = mc^2 = (2 \times 10^{-9})(3.00 \times 10^8)^2 = 1.8 \times 10^8$ J.

P11. Define potential energy.

Answer

Energy associated with the configuration of interacting objects or the position of a system relative to a chosen reference.

P12. Define gravitational potential energy near Earth's surface.

Answer

For a constant gravitational field, GPE = mgh relative to the selected zero height.

P13. What is elastic potential energy?

Answer

Energy stored when an elastic object is deformed; for an ideal spring it is $\frac{1}{2}kx^2$ relative to equilibrium.

P14. Explain the meaning of rest energy.

Answer

Mass itself corresponds to energy $E = mc^2$ even when an object is at rest.

P15. Relate work done by gravity to the change in gravitational potential energy.

Answer

$W_{gravity} = -\Delta GPE$. When an object falls, GPE decreases and gravity does positive work.

P16. A 6 kg object moves at 3.0 m/s. Find its translational kinetic energy.

Answer

$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 6 \times 3.0^2 = 27$ J.

P17. A 8 kg object moves at 3.0 m/s. Find its translational kinetic energy.

Answer

$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 8 \times 3.0^2 = 36$ J.

P18. A 6 kg object is 3.0 m above the chosen zero level. Find its gravitational potential energy. Use $g = 9.8\ m/s^2$.

Answer

$GPE = mgh = 6 \times 9.8 \times 3.0 = 176.4$ J.

P19. A 8 kg object is 3.0 m above the chosen zero level. Find its gravitational potential energy. Use $g = 9.8\ m/s^2$.

Answer

$GPE = mgh = 8 \times 9.8 \times 3.0 = 235.2$ J.

P20. A rotor has moment of inertia 0.6 kg m² and angular speed 5.0 rad/s. Find its rotational kinetic energy.

Answer

$KE_{rot} = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 0.6 \times 5.0^2 = 7.5$ J.

P21. A rotor has moment of inertia 0.8 kg m² and angular speed 5.0 rad/s. Find its rotational kinetic energy.

Answer

$KE_{rot} = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 0.8 \times 5.0^2 = 10$ J.

P22. A spring with constant 300 N/m is stretched 0.100 m from equilibrium. Find its elastic potential energy using $PE = \frac{1}{2}kx^2$.

Answer

$PE = \frac{1}{2}kx^2 = \frac{1}{2} \times 300 \times 0.100^2 = 1.5$ J.

P23. A spring with constant 400 N/m is stretched 0.100 m from equilibrium. Find its elastic potential energy using $PE = \frac{1}{2}kx^2$.

Answer

$PE = \frac{1}{2}kx^2 = \frac{1}{2} \times 400 \times 0.100^2 = 2$ J.

P24. Find the rest energy of mass 3×10^{-9} kg. Use $E = mc^2$ and $c = 3.00 \times 10^8$ m/s.

Answer

$E = mc^2 = (3 \times 10^{-9})(3.00 \times 10^8)^2 = 2.7 \times 10^8$ J.

P25. Find the rest energy of mass 4×10^{-9} kg. Use $E = mc^2$ and $c = 3.00 \times 10^8$ m/s.

Answer

$E = mc^2 = (4 \times 10^{-9})(3.00 \times 10^8)^2 = 3.6 \times 10^8$ J.

P26. A 2.0 kg object moves horizontally at 4.0 m/s at a height 3.0 m above the reference. Find KE and GPE.

Answer

$KE = \frac{1}{2} \times 2.0 \times 4.0^2 = 16$ J. $GPE = 2.0 \times 9.8 \times 3.0 = 58.8$ J.

P27. An object moves from height 5.0 m to 2.0 m. Explain the signs of ΔGPE and gravitational work.

Answer

$\Delta GPE = mg(2 - 5)$ is negative; $W_{gravity} = -\Delta GPE$ is positive. Gravity transfers energy as the object descends.

P28. If the GPE zero level is moved upward, does the physical gravitational work between two fixed heights change?

Answer

No. Both potential energies shift by the same amount, so ΔGPE and the work remain unchanged.

P29. Compare the energy changes when a spring's extension doubles and when an object's speed doubles.

Answer

Ideal spring potential energy and translational kinetic energy each increase by a factor of four because both depend on the square of the relevant variable.

P30. A wheel rolls without slipping. Why can its kinetic energy have two terms?

Answer

Its centre of mass translates while the wheel rotates about that centre. Total $KE = \frac{1}{2}mv_{CM}^2 + \frac{1}{2}I_{CM}\omega^2$ in the usual rigid-body description.